

Literature Survey

Introduction

Several Machine Learning techniques have been applied in precision agriculture to improve crop recommendation accuracy.

Existing Approaches

Decision Tree

Simple and easy to interpret but may overfit.

Logistic Regression

Effective for classification with relatively simple relationships.

Random Forest

Provides high accuracy by combining multiple decision trees.

Support Vector Machine (SVM)

Works well for complex classification but requires careful parameter tuning.

K-Means Clustering

Groups similar agricultural conditions for pattern analysis.

Research Gap

Many existing systems:

- Use limited environmental parameters.
- Ignore seasonal variations.
- Lack user-friendly interfaces.
- Do not provide real-time recommendations.

Proposed Improvement

OptiCrop combines Machine Learning with a Flask-based web application to provide accurate, fast, and accessible crop recommendations.